A Calibrated Trace-Driven 5G QoE Digital Twin for Trustworthy Next-Observation Forecasting and Grounded Explanations

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Abstract—This paper presents and audits a scientifically scoped, trace-driven Digital Twin for video-streaming Quality of Experience (QoE) forecasting in 5G New Radio. The prototype chronologically replays public 5G-QoERA observations, maintains network and service state, predicts whether the next observation has poor QoE, calibrates the resulting probability, and attaches a heuristic trust indicator with selective abstention. A persistence baseline and a network-only Logistic Regression are compared with a cross-layer Random Forest that combines throughput, location, resource allocation, packet loss, bitrate, resolution, current Mean Opinion Score (MOS), and past-only temporal features. The event is defined explicitly as $\mathrm{MOS}_{t+1} < 3.0$; the observed one-step interval has median 9.999 s. On 179,328 sealed chronological test rows, isotonic calibration and a validation-selected decision threshold of 0.37 yielded F1=0.8807, PR-AUC=0.9549, Brier=0.0715, and ECE=0.0116. There were 8,365 false alarms (7.96%). A validation-selected trust threshold abstained on 9.60% of cases and increased accepted-decision accuracy from 0.9018 to 0.9340 at 90.40% coverage. Finally, a grounded language-model component verbalizes a restricted evidence object but cannot change the prediction. All 30 retained explanations passed strict schema and deterministic grounding validation; one reviewer marked all 30 factually correct and complete with no unsupported statements. The results answer the research question affirmatively but with important qualifications: strong classification, low aggregate calibration error, and useful selective accuracy coexist in this dataset, while generalization, sample provenance, semantic grounding, and multi-reviewer evidence remain open limitations.

Index Terms—5G New Radio, Quality of Experience, digital twin, video streaming, chronological replay, probability calibration, Brier score, selective prediction, trustworthy artificial intelligence, grounded language model.

I. Introduction

Large populations of connected objects increasingly collect and transport short video segments through access networks and edge services. An airport camera, agricultural sensor, vehicle, or mobile terminal observes only a small part of a distributed service, yet the final user judges the assembled experience. Throughput alone cannot represent that judgment: packet loss, requested bitrate, resolution, scheduling resources, location, mobility, and recent QoE interact over time. A useful management prototype must therefore represent evolving cross-layer state, anticipate a service event, and report how much confidence an operator should place in the warning.

The public 5G-QoERA dataset was designed for integrated QoE analysis in 5G NR and combines user mobility, real base-station deployments, radio-map context, scheduling decisions, application settings, packet-loss observations, and MOS labels [1]. The dataset enables a small but scientifically valid trace-driven twin: a replay engine can consume recorded observations in time order, update state as an online service would, and test a future-event forecast without claiming to be a complete commercial twin.

This study asks: does a cross-layer model that classifies future poor QoE accurately also produce probabilities reliable enough for trustworthy network-management support? We operationalize "trustworthy" narrowly and measurably. The system must preserve chronology, prevent target leakage, calibrate its best classifier, quantify false alarms and timing, abstain when a validation-selected reliability score is too low, and reject a language-model explanation if it invents evidence.

The contributions are:

- a chronological, session-aware replay state spanning radio/resource, mobility, and video-service variables;
- a one-observation-ahead MOS event and leakage-safe temporal evaluation;
- the required comparison of persistence, network-only Logistic Regression, and cross-layer Random Forest models;
- validation-only calibration, threshold selection, trust scoring, and selective abstention;
- a schema-constrained, evidence-grounded explanation layer that is separated from prediction and control; and
- a reproducibility audit that regenerates every number in this report from authoritative frozen JSON rather than the superseded submission.

II. State of the Art

A. 5G Video QoE Modelling

QoE estimates the quality perceived by a user, whereas network Quality of Service describes technical delivery. For video, identical throughput can produce different experience under different requested bitrates; packet loss may have little effect in one operating region and severe effect in another. The novelty of 5G-QoERA is precisely this integration of mobility, scheduling, deployment, and application details with MOS [1]. Many conventional supervised pipelines nevertheless flatten such traces into independent rows. That practice overstates operational realism because a prediction at time t is conditioned by earlier state and because random splits can expose future operating regimes during training.

B. Trace-Driven Network Digital Twins

ITU-T Y.3090 specifies requirements and architecture for Digital Twin Networks [2]. The broader research literature describes a network twin as a data-driven representation that supports analysis, diagnosis, simulation, and management while remaining synchronized with the physical system [3]. The present work implements the limited offline subset appropriate to an assignment prototype: chronological trace replay, explicit state, future prediction, and an uncertainty/trust output. It does not claim bidirectional synchronization, counterfactual simulation, or autonomous control.

C. Calibration and Selective Prediction

A classifier's discrimination and its probability reliability are different properties. Guo et al.showed why post-hoc calibration must be evaluated separately from accuracy [4]. Platt scaling fits a sigmoid mapping [5]; isotonic regression instead learns a monotonic non-parametric mapping. Proper scores such as the Brier score [6] penalize probabilistic error, while ECE summarizes the weighted absolute difference between confidence and empirical frequency across bins [7]. ECE is useful but depends on binning and should not be interpreted as a guarantee for every individual case.

Selective classification adds a reject option. Rather than forcing a decision for every row, a system can trade coverage for lower risk among accepted predictions [8], [9]. This is suitable for an operator-support twin: an abstention requests more evidence or human review, while an accepted forecast can support a bounded decision. The trust score used here is explicitly heuristic, not a calibrated probability that the classifier is correct.

D. Language Models, Explanations, and Grounding

Language models can transform structured telemetry into concise operator text, but fluent generation may include unsupported claims. Hallucination surveys emphasize that factuality requires controls beyond linguistic quality [10]. This project therefore assigns the LLM no model training or control role. It receives a small allow-listed evidence object, returns a strict typed structure, cites evidence keys for each likely cause, and passes through a deterministic grounding gate. This is closer to a controlled verbalization layer than an autonomous diagnostic agent.

III. Work Description and Research Design

Figure 1 shows the complete prototype. Observations are reconstructed into sessions and replayed chronologically. Present and past-only features feed three classifiers. Validation data alone fits calibrators and selects operating thresholds. The test period is opened once for frozen evaluation. The chosen probability and trust score form a small evidence object. A language model proposes an explanation, which is retained only after schema and grounding validation.

The twin contains the four mandatory elements:

1) Replay: rows are globally sorted by timestamp and stable session identifier; per-session histories advance only after the current row is reached.
2) State: each session stores user, timestamp, base station, PRB, latitude, longitude, throughput, bitrate, resolution, packet loss, current MOS, session step, and elapsed session time.
3) Future event: the classifier predicts poor QoE at the next valid observation in the same session.
4) Reliability: a calibrated probability is accompanied by a trust score, level, and abstention decision.

The study is independent and trace-driven. A row represents one bitrate alternative attached to a physical radio observation; the prototype studies 248,624 physical observations and 97,248 reconstructed sessions rather than claiming that the dataset contains millions of distinct physical devices. This distinction matters when interpreting statistical independence and scale.

IV. Methodology

A. Dataset Preparation and Sampling

The upstream repository contains 32 tab-separated scenarios: four base stations by eight PRB allocations (5 through 40). The raw wide tables contain 4,147,536 physical rows. Four configured video alternatives (2,000, 4,000, 6,000, and 8,000 kbps) yield approximately 16.59 million long-format rows.

The frozen experiment uses a resource-safe, earliest-time group-balanced sample of 994,496 rows. It retains complete sessions and covers every base-station, PRB, and selected-bitrate group. The sample is authenticated by SHA-256, checked against all 32 trusted raw files, and then stripped of inherited session and target fields. Sessions and labels are reconstructed from base measurements. This audit found zero missing source rows, zero value mismatches in throughput, position, packet loss, or MOS, and zero duplicate row identities. After removing 97,248 terminal rows and 192 labels that cross chronological boundaries, 897,056 labelled rows remain.

Table I reports the global 60/20/20 time split. This evaluates temporal generalization, not generalization to unseen users or cells. It is important that the poor-QoE prevalence rises from 30.18% in training to 41.43% in test; accuracy alone would therefore hide a meaningful temporal distribution shift.

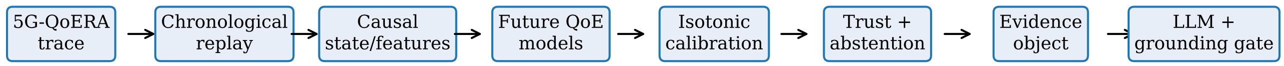

Fig. 1. Audited architecture. The language model is downstream of the prediction and trust decision and cannot modify either.

TABLE I
Chronological dataset partitions

Split	Rows	Sessions	Users	Poor rate
Train	537,504	51,232	216	30.18%
Validation	180,224	17,152	167	37.35%
Test	179,328	18,144	181	41.43%

B. Session Construction, Horizon, and Target

A trajectory is grouped by base station, PRB, bitrate, and user. A new session begins when the time gap exceeds 30 s. Within a session, the event is

$$y_t = \mathbb{1}[\mathrm{MOS}_{t+1} < 3.0]\,. \tag{1}$$

MOS exactly equal to 3.0 is not labelled poor. The last row of each session has no future target and is excluded. Labels whose $t+1$ timestamp crosses a split boundary are also excluded. The horizon is one observation, consistent with the trace sampling process. It is not a fixed 10-second horizon: the mean is 9.649 s, median 9.999 s, and 95th percentile 13.088 s. Equation 1 predicts future poor state, not the onset of degradation; poor-to-poor transitions are positive events.

C. Causal Feature Contract and Twin State

Feature generation sorts each session chronologically. Lag variables use shift; rolling means and standard deviations shift before rolling, so the historical window contains only times earlier than t. Current state is permitted because it is available at the prediction instant. Future MOS, future timestamp, target, and retrospective lead time are never model inputs or LLM evidence.

The network-only contract contains throughput, PRB, base station, position, time gap, past throughput, movement distance/speed, session step, and elapsed time. The cross-layer contract adds bitrate, resolution, current packet loss, current MOS, lagged and rolling MOS/loss, throughput-to-bitrate ratio, capacity margin, and an insufficiency flag. Consequently, the required radio, mobility, resource-allocation, video, and current-QoE aspects are represented.

TABLE II
Validation-selected calibration and thresholds

Model	Method	Threshold	F1	Brier	ECE
Network Logistic	Isotonic	0.35	0.6670	0.1886	0.0124
Cross-layer RF	Isotonic	0.37	0.8680	0.0741	0.0121

Replay parity is tested on interleaved synthetic sessions: the feature vector computed by online-style replay must equal the batch past-only vector. A future-invariance test perturbs later rows and verifies that earlier features and predictions do not change. These tests validate the implementation contract, although they do not replay every one of the 179,328 test rows at unit-test time.

D. Models

All learned preprocessing is fitted on training rows only.

- Persistence: predict future poor QoE when current MOS is below 3.0. Its hard 0/1 output supplies a strong temporal baseline.
- Network Logistic Regression: a balanced, standardized linear pipeline receives only the network feature contract.
- Cross-layer Random Forest: 100 trees, depth 14, minimum leaf size 20, 50% per-tree sampling, and balanced subsampling receive the full cross-layer contract; seed 42 fixes stochastic choices.

The comparison separates the value of cross-layer state from persistence and from a network-only linear model. No test labels are used in fitting, calibration, or threshold selection.

E. Calibration and Operating-Point Selection

The validation period is divided chronologically. The first 90,208 rows fit sigmoid and isotonic calibrators. After removing labels crossing the internal boundary, the later 89,952 rows select the calibration method by minimum ECE, then minimum Brier score, then maximum F1. A fixed threshold grid from 0.05 to 0.95 selects maximum F1, breaking ties by precision. Table II shows the frozen validation choices. Both learned models selected isotonic calibration; the Random Forest threshold was 0.37.

For test labels y_i and probabilities p_i, the Brier score is

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2, \tag{2}$$

and 10-bin ECE is

$$\text{ECE} = \sum_{b=1}^{B}\frac{|S_b|}{n}\left|\text{acc}(S_b) - \text{conf}(S_b)\right|. \tag{3}$$

We additionally report F1, precision, recall, PR-AUC, ROC-AUC, false-positive count, false-alarm rate FP/(FP+TN), and the observed next-row interval.

F. Trust Score and Abstention

Let τ be the selected classification threshold. Probability confidence is normalized distance from τ, reaching zero at the boundary and one at the endpoint on the corresponding decision side. The YAML-backed score is

$$T = 0.40C_\tau(p) + 0.30F_{1,\text{val}} \tag{4}$$

$$+ 0.20(1 - \text{ECE}_\text{val}) + 0.10Q_\text{data}. \tag{5}$$

Prediction stability is retained as a diagnostic field but, by design, has no weight in Equation 5. The trust score is a ranking heuristic, not $P(\text{correct})$. On the later validation half, the abstention threshold minimizes selective risk subject to at least 90% coverage. The frozen value 0.671185 achieves 90.20% validation coverage and is then applied unchanged to test.

G. Grounded LLM Component

The allow list contains decision, calibrated poor-QoE probability, trust score and level, current MOS, throughput, bitrate, capacity margin, throughput-to-bitrate ratio, and packet loss. It excludes labels and all future-derived fields. The requested output contains predicted event, confidence, summary, likely causes with exact evidence keys, operator checks, and limitations. For an abstention, confidence must be low, likely causes must be empty, and the text must say that evidence is insufficient.

Validation occurs in two stages. A strict Pydantic schema forbids extra or missing structure. A deterministic grounding validator checks prediction polarity, confidence/trust agreement, unsupported numbers, field/value and unit attribution, cause citations, and abstention behavior. Failed output is rejected rather than repaired silently. The external completion model and request metadata for the retained offline batch were not preserved, so this report deliberately makes no provider/model attribution.

V. Actions Performed and Software Implementation

The implementation work proceeded through the following auditable actions:

1) inspected the 32 public scenarios and normalized selected application bitrate/MOS columns into long form;
2) authenticated the pinned resource-safe sample and verified every sampled base measurement against source TSV bytes;
3) rebuilt sessions, next-row labels, and chronological split boundaries, discarding inherited derived fields;
4) implemented past-only features and a per-session replay engine with explicit state;
5) trained the three required models and stored model/configuration lineage;
6) fitted sigmoid and isotonic calibration on the first validation half and selected calibration, decision, and abstention thresholds on the second;
7) evaluated the frozen models once on the sealed test period;
8) exported 30 balanced explanation cases (10 accepted poor-QoE, 10 accepted acceptable-QoE, and 10 abstentions), validated completions, and completed one-person manual review;
9) added unit, causality, parity, lineage, grounding, and artifact-contract tests; and
10) replaced stale publication tables with a generator that checks five authoritative result documents before producing this report.

Source modules and scripts include docstrings, type hints, and focused comments around temporal shifts, split boundaries, trust semantics, evidence allow lists, and validation rules. The code is therefore submitted as the required commented implementation rather than copied into the paper. The repository's test suite exercises target generation, chronological splitting, replay, features, calibration, trust, models, evidence, explanation schema, grounding, human review, and artifact lineage.

VI. Results

A. Model Comparison

Table III evaluates all models on the same 179,328 test rows. The persistence baseline is strong (F1 0.8403), confirming high temporal autocorrelation in MOS. The network-only Logistic Regression underperforms both alternatives, particularly in false alarms. The cross-layer Random Forest achieves the best F1 and PR-AUC and the lowest Brier score and ECE.

The final Random Forest yields accuracy 0.9018, precision 0.8860, recall 0.8755, F1 0.8807, PR-AUC 0.9549, ROC-AUC 0.9637, Brier 0.0715, and ECE 0.0116. Its confusion matrix contains 96,671 true negatives, 8,365 false positives, 9,250 false negatives, and 65,042 true positives. Thus the false-alarm rate is 7.96% and the miss rate is 12.45%. Reporting both counts and rates avoids hiding the operational burden created by a large test set.

The model comparison supports the cross-layer hypothesis. The Random Forest improves F1 by 0.0405 over persistence and by 0.2023 over network-only Logistic Regression. Its PR-AUC exceeds persistence by 0.1830 and the network-only model by 0.3090. Because the test

TABLE III
Sealed-test comparison; lower Brier, ECE, and false-alarm rate are better

Model	F1	PR-AUC	Brier	ECE	FP	FAR
Persistence	0.8403	0.7719	0.1325	0.1325	11,951	11.38%
Network Logistic	0.6784	0.6459	0.1941	0.0216	45,512	43.33%
Cross-layer RF	0.8807	0.9549	0.0715	0.0116	8,365	7.96%

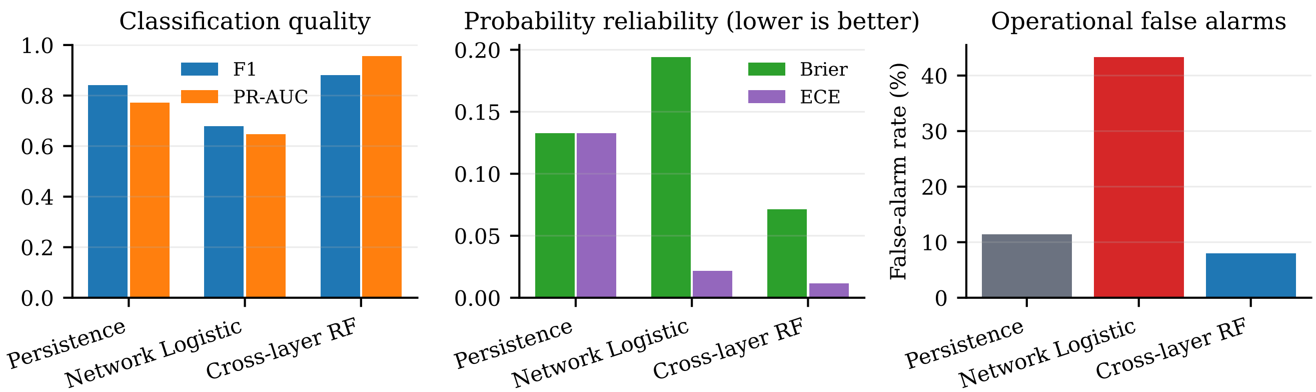

Fig. 2. Classification, probability-reliability, and false-alarm comparison on the sealed test period.

positive rate is 41.43%, PR-AUC is more informative than accuracy alone.

B. Calibration and the Research Question

Low ECE and Brier error show that the best classifier is also well calibrated in aggregate on this chronological test period. This is not automatic: the network-only model has ECE 0.0216 but poor F1 0.6784, whereas persistence has strong F1 but ECE 0.1325 because its binary scores are overconfident. Discrimination and calibration must therefore be measured separately. The cross-layer Random Forest is the only model that combines high PR-AUC with low Brier and ECE.

The answer to the main research question is consequently yes for the frozen 5G-QoERA time period: the best classification model is sufficiently calibrated to support a selective operator policy. The answer is not a claim of universal reliability; ECE is aggregate and later sections identify drift, sampling, and external-validity limits.

C. Trust, Coverage, and Selective Accuracy

Applying the validation-selected threshold abstains on 17,212 of 179,328 rows (9.60%) and covers 90.40%. Table IV and Figure 3 show the resulting trade-off. Accepted accuracy rises from 0.9018 to 0.9340; accepted precision is 0.9299, recall 0.9091, and F1 0.9194. Accepted Brier score and ECE also improve to 0.0542 and 0.0090.

This improvement is descriptive, not a second test-tuned operating point: the trust policy and threshold were fixed entirely on validation data. Coverage remains above the 90% constraint. An operator can interpret an accepted high/medium-trust decision as eligible for support, while

TABLE IV
Overall versus accepted test decisions

Scope	Acc.	Prec.	Recall	F1	Brier	ECE
All predictions	0.9018	0.8860	0.8755	0.8807	0.0715	0.0116
Accepted only	0.9340	0.9299	0.9091	0.9194	0.0542	0.0090

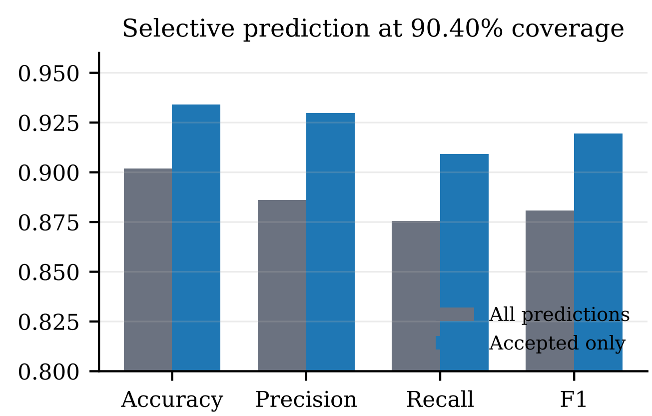

Fig. 3. Abstention improves accepted-decision performance while retaining 90.40% coverage.

an abstention means “collect or verify more evidence,” not “acceptable QoE.”

D. Cross-Layer Drivers

Figure 4 ranks the leading impurity-based Random Forest features. Packet-loss percentage and throughput-to-bitrate ratio dominate, followed by capacity margin,

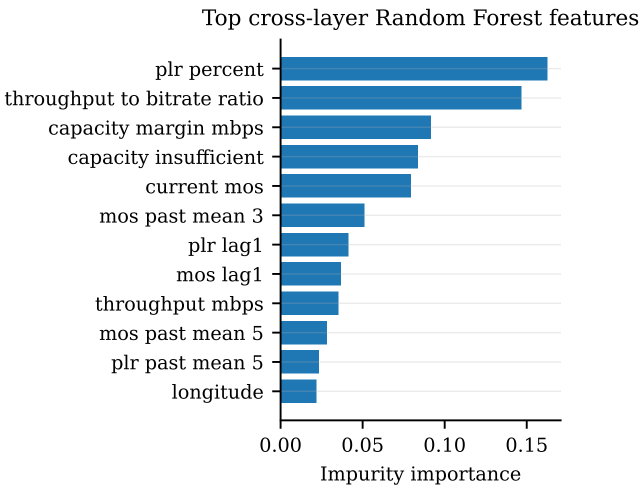

Fig. 4. Top twelve cross-layer Random Forest features.

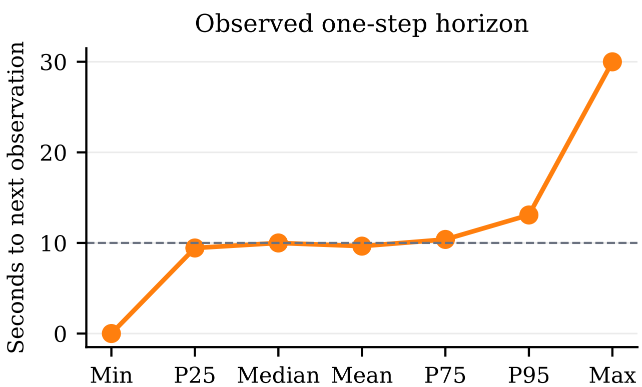

Fig. 5. Observed time from each prediction row to its next in-session row.

capacity insufficiency, and current MOS. Past MOS and loss variables also contribute. This pattern is consistent with the research design: future experience depends on delivery capacity relative to application demand and on recent service state, not on throughput in isolation. Importance is associative and global; it does not prove a causal network fault for an individual event.

E. Prediction Horizon

Figure 5 summarizes the actual one-step interval. Most rows are near 10 s, but the range extends from roughly zero to 30 s. The median 9.999 s provides a short investigation window for an edge operator, yet it should not be reported as guaranteed "lead time before onset." The target is next-state poor QoE, and the dataset does not label a unique degradation onset.

VII. Explanation Evaluation

The balanced 30-case batch contains the three operational outcomes needed to exercise the interface. Table V

TABLE V
Grounded explanation evaluation

Measure	Result
Retained/reviewed	30/30
Schema valid	30 (100%)
Grounding valid	30 (100%)
Factual and complete	30 (100%)
Unsupported statements	0 (0%)
Contradictions	0 (0%)

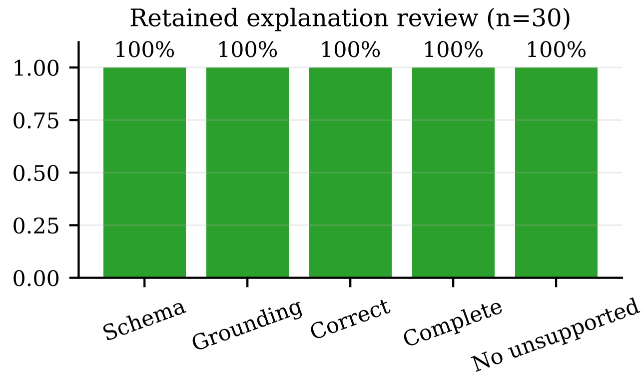

Fig. 6. Validation and manual-review outcomes for the retained explanation batch.

and Figure 6 report the retained evaluation. All outputs parse under the strict schema and pass the grounding validator. The single reviewer judged all 30 factually correct and complete, with zero unsupported-statement and contradiction flags.

A typical accepted poor-QoE explanation reports the future event and confidence, then associates a negative capacity margin and packet loss with the forecast, citing the corresponding keys. It tells the operator to verify throughput against bitrate, inspect packet loss, and confirm whether current MOS conditions persist. It also states that evidence does not identify a unique root cause. For an abstention, the explanation makes no causal diagnosis and recommends collecting another observation. This satisfies the requirement to explain the event, likely causes, confidence/limitations, and operator checks while preventing invented numeric values.

The 100% rates must be interpreted cautiously. The set is small, deliberately balanced rather than prevalence weighted, reviewed by one person, and based on externally completed outputs whose provider/model metadata were not retained. The result demonstrates a working grounded interface, not population-wide LLM quality or inter-rater reliability.

VIII. Discussion

Three findings matter for trustworthy management. First, persistence is a demanding baseline. A future-QoE paper that omitted it could mistake temporal autocorrelation for model intelligence. The cross-layer model nevertheless adds measurable value in F1, PR-AUC, and false alarms.

Second, classification and calibration can disagree. Network-only Logistic Regression is relatively calibrated yet not sufficiently discriminative at the chosen operating point; persistence classifies well but supplies poor probabilities. The calibrated cross-layer Random Forest performs well on both dimensions. This directly answers the study's central question and justifies reporting F1, PR-AUC, Brier, and ECE together.

Third, a single probability does not complete a trustworthy interface. The validation-selected reject option improves the accuracy of decisions offered to an operator and explicitly routes ambiguous cases to further verification. The explanation layer then makes evidence and limitations visible without controlling the predictor. This separation of authority is valuable: model output, reliability policy, and language generation can each fail independently and are validated at their boundaries.

The project also illustrates scale honestly. Millions of modelling rows and tens of thousands of sessions exercise distributed trace processing, but they do not constitute millions of independent physical Things. Four bitrate alternatives share each physical radio observation, and repeated rows from a user/session are correlated. The prototype addresses the assignment's large-scale management context without inflating its empirical unit count.

IX. Limitations and Threats to Validity

Offline scope: trace replay approximates online evolution but cannot reproduce feedback between a control action and future radio conditions. No closed-loop resource change is executed.

Sampling and provenance: the final experiment uses a pinned, earliest-time group-balanced sample rather than the complete 16.59-million-row long table. Raw compatibility and byte identity are strong, but the historical command that produced the sample is unavailable. It should therefore remain labelled a quarantined external snapshot. The result is scientifically auditable but not fully regenerable from the public raw data alone without a new declared sampling policy and new results.

Dependence and generalization: bitrate alternatives share radio observations, and sessions/users may recur across periods. A global time split tests later-time performance, not unseen-user, unseen-cell, or unseen-city performance. Results may not transfer to other codecs, services, deployments, or traffic regimes.

Target construct: MOS below 3.0 is an explicit, reproducible poor-state definition, but other thresholds or onset-only targets would change prevalence and action meaning. The one-step interval is irregular and short.

Calibration: ECE depends on binning and aggregate test performance can hide subgroup miscalibration. Isotonic mappings may require recalibration under drift. The trust equation is a policy heuristic; its weights were not learned from measured operational cost.

Interpretation: impurity importance and LLM likely-cause statements express supported associations, not causality. Automatic grounding focuses on structured fields, values, units, and obvious contradictions; it cannot prove that every non-numeric semantic implication is warranted.

Explanation evaluation: 30 cases and one reviewer do not establish inter-rater reliability. Missing external completion metadata also limits exact generation reproducibility, even though the retained outputs, prompts, schemas, validations, and human judgments are preserved.

X. Suggestions for Improvement

The following actions would materially strengthen the work:

1) publish a deterministic sample builder or rerun the full dataset on higher-memory infrastructure, recording data, configuration, code, model, and result hashes in one run manifest;
2) evaluate rolling-origin and blocked cross-validation, plus held-out users, cells, and scenario files, to separate temporal from entity generalization;
3) compare onset-specific, fixed-time, and multi-horizon targets and report event-based warning lead time rather than only the next-row interval;
4) add subgroup reliability diagrams, adaptive/calibration error variants, class-conditional calibration, bootstrap confidence intervals, and drift monitoring;
5) choose trust and decision thresholds from explicit false-alarm, miss, abstention, and intervention costs, then validate the policy prospectively;
6) compare calibrated Gradient Boosting and temporal models under the same leakage-safe protocol, with ablations for application, mobility, and historical QoE feature families;
7) extend the twin to streaming ingestion and counterfactual "what-if" resource-allocation experiments, while keeping the LLM outside the control loop;
8) strengthen grounding with semantic entailment checks, unit ontologies, allow-listed causal templates, adversarial prompts, and rejection audits; and
9) retain provider/model/version/request metadata and conduct blinded review with multiple network-domain operators, reporting inter-rater agreement and usefulness/actionability scores.

XI. Requirement Compliance Audit

Table VI maps the assignment to implemented, testable evidence. All mandatory prototype and report elements pass. The raw-to-final rerun limitation is disclosed as a reproducibility improvement rather than hidden.

TABLE VI
Final requirement audit

Requirement	Audited evidence	Status
IEEE paper, ≤15 pages excluding references	IEEEtran two-column source; state of the art, work, methodology, actions, results, limitations, and improvements are explicit; build-time page audit included	Pass
Commented code	Typed modules and scripts contain docstrings and comments for temporal, calibration, trust, evidence, and grounding logic	Pass
Chronological replay	Global timestamp/session ordering, per-session state, replay-parity and future-invariance tests	Pass
Network/service state	Throughput, base station, PRB, location/mobility, packet loss, bitrate, resolution, current MOS, and temporal state	Pass
Future event and horizon	$\mathrm{MOS}_{t+1} < 3.0$ within session; one-step interval reported with timing statistics	Pass
Required model comparison	Persistence, network-only Logistic Regression, and cross-layer Random Forest on identical test rows	Pass
Calibrated best model	Sigmoid/isotonic compared on validation; isotonic RF selected and frozen before test	Pass
Required evaluation	F1, PR-AUC, Brier, ECE, false alarms, and observed prediction interval all reported	Pass
Confidence/trust	Calibrated probability, heuristic trust score, levels, validation-selected abstention and coverage	Pass
Grounded LLM	Structured evidence, event/cause/confidence/limitations/operator checks, strict schema and deterministic no-invention validation	Pass

XII. Conclusion

This work delivers the requested small, scientifically valid 5G QoE Digital Twin. It chronologically replays the public 5G-QoERA trace, maintains a cross-layer state, forecasts next-observation poor QoE, calibrates its best model, quantifies trust and abstention, and produces evidence-grounded operator explanations. The calibrated Random Forest achieves F1 0.8807, PR-AUC 0.9549, Brier 0.0715, and ECE 0.0116 on 179,328 later-time rows. Selective abstention preserves 90.40% coverage while raising accepted accuracy to 0.9340. The 30 retained explanations pass automated and manual factuality checks without unsupported values.

The central conclusion is nuanced: good cross-layer classification and good aggregate calibration do coexist here, and a validation-selected reject option improves the reliability of operator-facing decisions. The prototype remains an offline research twin, not a commercial system. Full raw-data reproducibility, external scenario validation, cost-based trust design, drift monitoring, semantic grounding, and multi-reviewer explanation studies are the most important next steps.

Acknowledgment

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